Joseph T. Iosue

Doctoral Candidate

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Education

2020 – **University of Maryland**, *Doctor of Philosophy (Physics)*, JQI Graduate Fellow
Advisors: Prof. Alexey Gorshkov and Prof. Victor Albert

2015 – 2019 **Massachusetts Institute of Technology**, *Bachelor of Science*
B.S. (Physics), Minor (Computer Science) – GPA 4.9/5.0

Research Experience

2020 – **University of Maryland, Department of Physics**, *Research Assistant*
Affiliations:
- Joint Quantum Institute (JQI)
- Joint Center for Quantum Information and Computer Science (QuICS)

2023 **Johns Hopkins University Applied Physics Laboratory**, *Summer intern*
Supervisor: Dr. Paraj Titum. Studied a specific quantum sensing and noise characterization problem. Paper in progress.

2019 – 2020 **QC Ware, Corp**, *Quantum Algorithms Researcher*
Developed software and algorithms for customer use cases, including [1].

2018 **Los Alamos National Laboratory**, *Quantum Computing Summer Fellowship*
Supervisor: Dr. Patrick Coles. Developed and published a novel quantum algorithm [3].

2017 **Joint Quantum Institute, University of Maryland**, *Summer Researcher*
Supervisor: Prof. Alexey Gorshkov. Studied quantum phase transitions via quench dynamics [2].

2015 – 2018 **Massachusetts Institute of Technology**, *Undergraduate researcher*
- (2017 – 2018) Supervisor: Prof. Or Hen. Studied proton/neutron dynamics with C++ and ROOT.
- (2016) Supervisor: Prof. Nuno Loureiro. Modeled particle transport in turbulent media using C.
- (2015) Supervisor: Prof. Emilio Baglietto. Modeled nuclear waste storage canisters and fission waste.

Awards and Achievements

2023 **Boulder Summer School**, *Non-Equilibrium Quantum Dynamics*, University of Colorado Boulder
One of 81 accepted into the four-week program

2020 – 2025 **JQI Graduate Fellowship**, Joint Quantum Institute, University of Maryland

2019, 2020 **NSF GRFP Honorable Mention**, National Science Foundation

Service and Mentoring

Journal and Conference Reviews/Subreviews
- Quantum journal
- Quantum Information Processing (QIP) 2024, 2023

2023 **Mentor**
Mentored high school summer student Jason Youm studying extensions of my work [5]. Paper in progress.

2020 – 2022 **Volunteer Tutor**, *UMD Department of Physics*

2016 **Teaching assistant and grader**, *MIT Department of Physics*

Publications

See also: Google Scholar, arXiv

[9] **J. T. Iosue**, K. Sharma, M. J. Gullans, and V. V. Albert, "Continuous-variable quantum state designs: theory and applications", Phys. Rev. X **14**, 011013 (2024), 10.48550/arXiv.2211.05127 (2022).

[8] Z. Liang, Y. Xu, **J. T. Iosue**, and Y.-A. Chen, "Extracting topological orders of generalized Pauli stabilizer codes in two dimensions", 10.48550/arXiv.2312.11170 (2023).

[7] A. Ehrenberg, **J. T. Iosue**, A. Deshpande, D. Hangleiter, and A. V. Gorshkov, "Transition of anticoncentration in Gaussian Boson Sampling", 10.48550/arXiv.2312.08433 (2023).

[6] **J. T. Iosue**, T. C. Mooney, A. Ehrenberg, and A. V. Gorshkov, "Projective toric designs, difference sets, and quantum state designs", 10.48550/arXiv.2311.13479 (2023).

[5] **J. T. Iosue**, A. Ehrenberg, D. Hangleiter, A. Deshpande, and A. V. Gorshkov, "Page curves and typical entanglement in linear optics", Quantum **7**, 1017 (2023), 10.48550/arXiv.2209.06838 (2022).

[4] S. P. Jain, **J. T. Iosue**, A. Barg, and V. V. Albert, "Quantum spherical codes", 10.48550/arXiv.2302.11593 (2023).

[3] C. Cirstoiu, Z. Holmes, **J. T. Iosue**, L. Cincio, P. J. Coles, and A. Sornborger, "Variational fast forwarding for quantum simulation beyond the coherence time", npj Quantum Information **6**, 82 (2020), 10.48550/arXiv.1910.04292 (2019).

[2] P. Titum, **J. T. Iosue**, J. R. Garrison, A. V. Gorshkov, and Z.-X. Gong, "Probing ground-state phase transitions through quench dynamics", Physical Review Letters **123**, 115701 (2019), 10.48550/arXiv.1809.06377 (2018).

[1] R. M. Parrish, **J. T. Iosue**, A. Ozaeta, and P. L. McMahon, "A Jacobi diagonalization and Anderson acceleration algorithm for variational quantum algorithm parameter optimization", 10.48550/arXiv.1904.03206 (2019).

Talks

Projective toric designs, difference sets, and quantum state designs, *based on [6]*

- 2024 (Scheduled) – Codes and Expansions (CodEx) Seminar (invited, virtual), *see recording on YouTube*

Continuous-variable quantum state designs: theory and applications, *based on [9]*

- 2023 – Quantum Information Processing (QIP), *see recording on YouTube*
- 2023 – APS March Meeting
- 2022 – Prof. David Gross's group seminar (invited, virtual)
- 2022 – CU Boulder journal club (invited, virtual)
- 2022 – University of Maryland JQI-QuICS quantum seminar
- 2022 – APS March Meeting (virtual)

Page curves and typical entanglement in linear optics, *based on [5]*

- 2023 – Quantum Algorithms and Applications Collaboratory (QuAAC) Seminar, Sandia National Laboratory (invited, given by coauthor)
- 2023 – APS March Meeting

Posters

Page curves and typical entanglement in linear optics, *based on [5]*

- 2023 – Boulder Summer School
- 2023 – Quantum Information Processing (QIP)

An initial condition robust outer-loop optimization strategy for QAOA

- 2019 – TQC Conference, College Park, Maryland

Selected Projects

2019 – **qubovert**, *Python package (with C extension) for binary optimization*

- Created qubovert, which is particularly designed to aid in converting optimization problems to a form that can be solved with quantum annealers and quantum optimization algorithms.
- qubovert can be installed with `pip install qubovert`, the source code is hosted at `github.com/jtiosue/qubovert`, and the documentation is hosted at `qubovert.readthedocs.io`.
- qubovert currently has **over 281k downloads** from PyPI and 34 stars and 8 forks on GitHub.

2019 **Powell bounded multivariate optimization**, *SciPy contribution*

- Authored pull request number 10648 on Python's SciPy package. My contribution is included in the 1.5.0 release and later releases
- The pull request implements an additional feature for SciPy's minimization method. I devised a bounded version of the standard unbounded Powell minimization method and found it to often perform much better than the other gradient-free minimizers. I then implemented this variant in SciPy's software stack and created the pull request.

2018 **Quantum Computer Simulator**, *C++ project*

- Implemented a quantum computer simulator in C++.

2023 **rcal**, *Python package for review calibration*

- Devised a novel review calibration algorithm (written report)
- Implemented the algorithm in a Python package.

Miscellaneous

2023 **On-Line Encyclopedia of Integer Sequences**, *based on [6]*
Our work is included in the OEIS:A108625 entry

2022 **On-Line Encyclopedia of Integer Sequences**, *based on [5]*
Our work is included in the OEIS:A062991 entry

2020 **PHYS.ORG**, Popular science article, *based on [3]*
New quantum computing algorithm skips past time limits imposed by decoherence, Online

Skills

Programming Python, C, C++, Mathematica/WolframScript, Javascript, Julia, LaTeX, Git